

Minkowski sum of convex polygons

Definition

Consider two sets A and B of points on a plane. Minkowski sum $A + B$ is defined as $\{a + b \mid a \in A, b \in B\}$. Here we will consider the case when A and B consist of convex polygons P and Q with their interiors. Throughout this article we will identify polygons with ordered sequences of their vertices, so that notation like $|P|$ or P_i makes sense. It turns out that the sum of convex polygons P and Q is a convex polygon with at most $|P| + |Q|$ vertices.

Algorithm

Here we consider the polygons to be cyclically enumerated, i. e. $P_{|P|} = P_0$, $Q_{|Q|} = Q_0$ and so on.

Since the size of the sum is linear in terms of the sizes of initial polygons, we should aim at finding a linear-time algorithm. Suppose that both polygons are ordered counter-clockwise. Consider sequences of edges $\{\overrightarrow{P_i P_{i+1}}\}$ and $\{\overrightarrow{Q_j Q_{j+1}}\}$ ordered by polar angle. We claim that the sequence of edges of $P + Q$ can be obtained by merging these two sequences preserving polar angle order and replacing consecutive co-directed vectors with their sum. Straightforward usage of this idea results in a linear-time algorithm, however, restoring the vertices of $P + Q$ from the sequence of sides requires repeated addition of vectors, which may introduce unwanted precision issues if we're working with floating-point coordinates, so we will describe a slight modification of this idea.

Firstly we should reorder the vertices in such a way that the first vertex of each polygon has the lowest y-coordinate (in case of several such vertices pick the one with the smallest x-coordinate). After that the sides of both polygons will become sorted by polar angle, so there is no need to sort them manually. Now we create two pointers i (pointing to a vertex of P) and j (pointing to a vertex of Q), both initially set to 0. We repeat the following steps while $i < |P|$ or $j < |Q|$.

1. Append $P_i + Q_j$ to $P + Q$.